SONA ROSE DEVASIA

B.Tech Computer Science Student

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SUMMARY

B.Tech Computer Science student with hands-on experience in Python, data structures, backend development, databases, machine learning, and full-stack application development. Experienced in building practical software solutions using FastAPI, React, SQL, Docker, and other development tools through internships and projects. Adaptable and willing to learn new technologies while applying technical knowledge to real-world projects.

WORK EXPERIENCE

Data Science & Machine Learning Intern | NeST Digital, Kochi (*Feb 2026 - Mar 2026)*

- Preprocessed and analyzed real-world datasets using Python for machine learning tasks.
- Applied machine learning algorithms to build and evaluate predictive models.
- Visualized data insights and model results using Python libraries to support analysis and interpretation.

Machine Learning Intern | Infosys Springboard (Virtual) (*Oct 2025 - Dec 2025)*

- Developed an NLU-based chatbot training platform using spaCy, BERT, and Logistic Regression for model training and evaluation
- Built an end-to-end web application using Streamlit, FastAPI, and SQLite for data handling and user interaction
- Tested and validated model performance using standard evaluation metrics

PROJECTS

Nutrition Assistant Grocery Store Web App | *Python, HTML, CSS, JavaScript, FastAPI, SQLite*

- Developed a full-stack grocery platform with user authentication, cart/order management, and an admin panel for inventory control.
- Built a nutrition recommendation engine that analyzes cart contents against a balanced nutrient profile and suggests items to cover gaps (e.g., protein, vitamin, fiber).

Bot Trainer - NLU Model Trainer & Evaluator | *Streamlit, FastAPI, spaCy, BERT, Logistic Regression*

- Built an end-to-end NLU platform for uploading datasets, annotating intents/entities, and training models via spaCy, BERT, and Logistic Regression.
- Implemented model evaluation (accuracy, precision, recall, F1) with a version-comparison dashboard, and designed an active learning loop that flags low-confidence predictions for human review and retrains the model to improve accuracy over time.

Real-Time Infrastructure Monitoring Platform | *Python, FastAPI, React, Docker, Docker Compose*

- Developed a full-stack dashboard for real-time monitoring of CPU, RAM, disk usage and system uptime using FastAPI, psutil, React and Recharts.
- Containerized the frontend and backend using Docker and Docker Compose, optimized Dockerfile layer caching and enabled single-command application startup.

DFU-AI – Diabetic Foot Ulcer Classification & Severity Analysis | *Python, TensorFlow, CNN Feature Fusion*

- Developed a two-stage deep learning system using multi-CNN feature fusion (DenseNet201, VGG19, NASNetMobile) to classify foot images as Healthy or Diabetic Foot Ulcer (DFU), followed by severity classification into four grades.
- Implemented CNN fine-tuning, GAP-based feature fusion, model evaluation, and Explainable AI (Grad-CAM, SHAP, LIME) to interpret predictions, with an image-upload interface for classification and visualization.

LoRa-Based Landslide Early Warning System | *ESP32, LoRa SX1278, SPI, GPIO, Arduino IDE*

- Developed an IoT-based landslide early-warning system using ESP32 for environmental sensing and SX1278 LoRa for wireless transmission of sensor data.
- Implemented sensor data collection and wireless transmission using SPI communication and GPIO interrupts for real-time environmental monitoring.

EDUCATION

B.Tech - Computer Science and Engineering *2023 - 2027*
Amal Jyothi College of Engineering (Autonomous), Kanjirappally | **CGPA: 9.28 / 10.00**

Higher Secondary (Class XII - CBSE) *2023*
Little Flower Public School and Junior College, Kollamula | **88.8%**

Secondary (Class X - CBSE) *2021*
Little Flower Public School and Junior College, Kollamula | **94.8%**

SKILLS

Programming: Python, C, Java (Basics)

Web: HTML, CSS, JavaScript, React

Frameworks: Flask, FastAPI, Streamlit

Database: SQL, SQLite

Libraries: TensorFlow, Keras, Scikit-learn, spaCy, psutil

Tools: GitHub, VS Code, Figma, Google Colab, Docker, Docker Compose

Concepts: Data Structures, Machine Learning, DBMS

Soft Skills: Teamwork, Adaptability, Time Management, Quick Learning

CERTIFICATIONS

- Python for Data Science, AI & Development - IBM | Coursera | 2025
- Database Management Systems - Elite | NPTEL (IIT Kharagpur) | 2025
- Oracle Java Foundation Learner's | 2024
- HTML5, CSS3 & JavaScript | Infosys Springboard | 2025

ACHIEVEMENTS

- 3rd Prize - Cyfusion (Design Thinking & Pitching), SAHYA, Marian College Kuttikanam | 2026
- Qualified for Phase 2 - Innovation Ideas Unleashed (I2U), AJCE | 2025

REFERENCES

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